1 Grammar

$$
\begin{aligned}
\text{Program} &::= \text{Definition}^* \ \text{Term} \\
\text{Definition} &::= \texttt{let}\ \text{Var}\ \texttt{=}\ \text{Term} \\
\text{Term} &::= \text{Abstraction} \mid \text{Application} \mid \text{Atom} \\
\text{Abstraction} &::= \backslash\ \text{Vars}\ .\ \text{Term} \\
\text{Vars} &::= \text{Var}\ \text{Vars} \mid \text{Var} \\
\text{Application} &::= \text{Atom}\ \text{AppSuffix} \\
\text{AppSuffix} &::= \text{Atom}\ \text{AppSuffix} \mid \text{Atom} \\
\text{Atom} &::= \text{Var} \mid (\ \text{Term}\ ) \\
\text{Var} &::= [a - zA - Z]+
\end{aligned}
$$

2 Derivation Examples

2.1 Example 1: `\x.\y.x y`

$$
\begin{aligned}
&\text{Program} \\
&\rightarrow \text{Term} \\
&\rightarrow \text{Abstraction} \\
&\rightarrow \backslash\ \text{Vars}\ .\ \text{Term} \\
&\rightarrow \backslash\ \text{Var}\ .\ \text{Term} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \text{Term} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \text{Abstraction} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \text{Vars}\ .\ \text{Term} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \text{Var}\ .\ \text{Term} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \texttt{y}\ .\ \text{Term} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \texttt{y}\ .\ \text{Application} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \texttt{y}\ .\ \text{Atom}\ \text{AppSuffix} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \texttt{y}\ .\ \text{Var}\ \text{AppSuffix} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \texttt{y}\ .\ \texttt{x}\ \text{AppSuffix} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \texttt{y}\ .\ \texttt{x}\ \text{Atom} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \texttt{y}\ .\ \texttt{x}\ \text{Var} \\
&\rightarrow \backslash\ \texttt{x}\ .\ \backslash\ \texttt{y}\ .\ \texttt{x}\ \texttt{y} \\
&= \texttt{\textbackslash x.\textbackslash y.x y}
\end{aligned}
$$

2.2 Example 2: `let I = \x.x in I I`

Program
→ Definition Program
→ `let` Var = Term Program
→ `let I` = Term Program
→ `let I` = Abstraction Program
→ `let I` = `\` Vars `.` Term Program
→ `let I` = `\` Var `.` Term Program
→ `let I` = `\ x .` Term Program
→ `let I` = `\ x .` Atom Program
→ `let I` = `\ x .` Var Program
→ `let I` = `\ x . x` Program
→ `let I` = `\ x . x` Term
→ `let I` = `\ x . x` Application
→ `let I` = `\ x . x` Atom AppSuffix
→ `let I` = `\ x . x` Var AppSuffix
→ `let I` = `\ x . x I` AppSuffix
→ `let I` = `\ x . x I` Atom
→ `let I` = `\ x . x I` Var
→ `let I` = `\ x . x I I`
= `let I = \x.x I I`